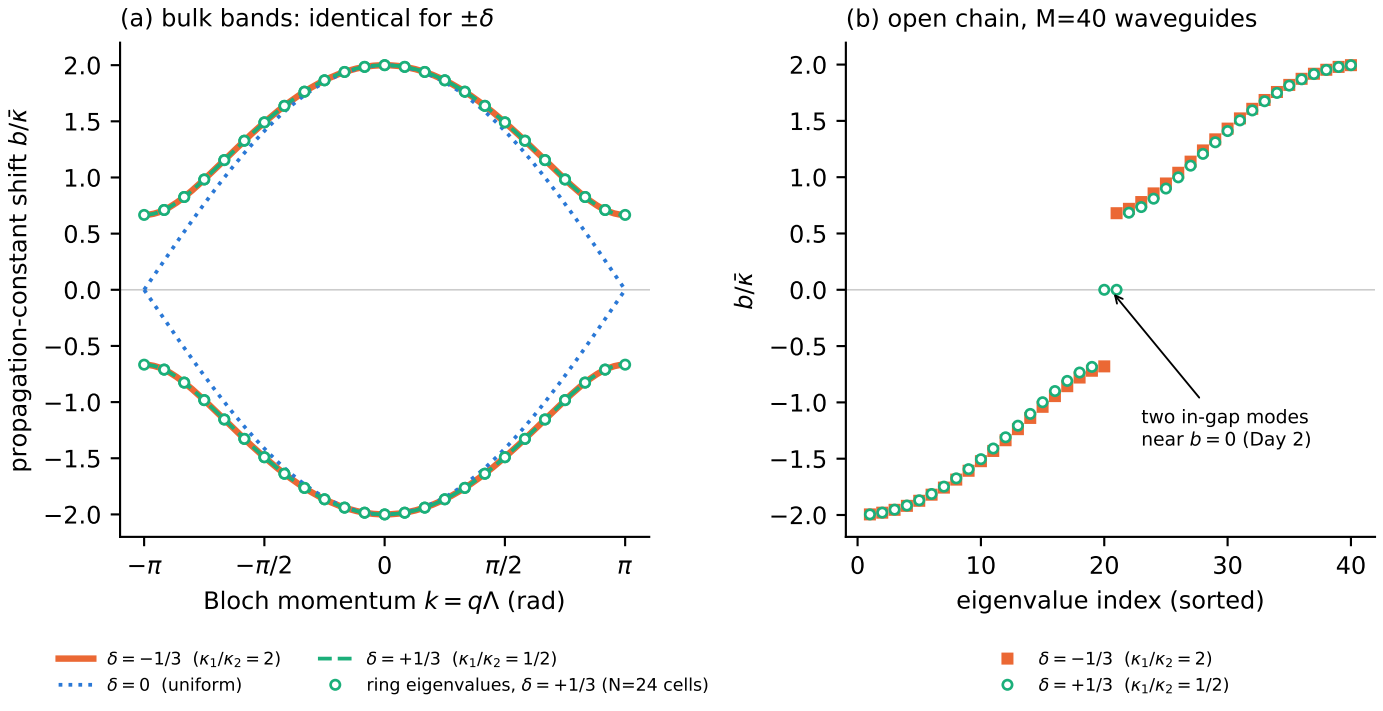
(a) bulk bands: identical for ±δ
propagation-constant shift b/κ̄
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−π
−π/2
0
π/2
π
Bloch momentum k = qΛ (rad)
δ = −1/3 (κ1/κ2 = 2)
δ = +1/3 (κ1/κ2 = 1/2)
δ = 0 (uniform)
ring eigenvalues, δ = +1/3 (N=24 cells)
(b) open chain, M=40 waveguides
b/κ̄
0
10
20
30
40
eigenvalue index (sorted)
two in-gap modes
near b = 0 (Day 2)
δ = −1/3 (κ1/κ2 = 2)
δ = +1/3 (κ1/κ2 = 1/2)